

Student Database Analysis

Project Overview

The Student Database Analysis project demonstrates the use of SQL to design and manage a simple student database. The project stores information about students, courses, marks, and attendance, and performs data analysis using SQL queries.

Objectives

- Design a relational student database.
 - Insert sample student records.
 - Perform filtering, sorting, and searching operations.
 - Apply SQL aggregate functions.
 - Generate reports for student performance.
 - Implement SQL JOIN operations.
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Database Design

The project consists of three tables:

- Students
- Courses
- Marks

The tables are connected using primary keys and foreign keys to maintain data integrity.

SQL Operations Performed

The following SQL concepts were implemented:

- SELECT
 - WHERE
 - ORDER BY
 - Aggregate Functions (COUNT, SUM, AVG, MAX, MIN)
 - INNER JOIN
 - Filtering and Sorting
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Analysis Performed

The following reports were generated:

- Student details
 - Course details
 - Marks obtained
 - Top-performing students
 - Attendance report
 - Average marks
 - Highest and lowest marks
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Results

The project successfully demonstrated how SQL can be used to organize and analyze student information. Reports generated using SQL help evaluate student performance and attendance efficiently.

Conclusion

This project strengthened the understanding of relational database design and basic SQL operations. It provides a solid foundation for data management and reporting using SQL.